

Botnets, and the CyberCriminal Underground

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Abstract—An underground community of cyber criminals has grown in recent years with powerful technologies capable of inflicting serious economic and infrastructural harm in the digital age. This paper serves as an introduction to the world of botnets and to the efforts of the nonprofit group "The ShadowServer Foundation" to track them. A data mining exploration is performed on ShadowServer's datasets to investigate possible classification mechanisms for threat assessment.

Index Terms— Botnets, SNA, clustering

I. BOTNETS

Botnets are the latest stage in the evolution of computer crime technology.

The earliest malware was typically used for pranks, either for damaging systems, or printing taunting messages to the user. Traditional computer viruses were engineered to self-copy themselves to other legitimate host programs on the system. These infected host files could then be carried to other computers by unknowing computer users, or by the underlying infected programs. Trojan horses arose to infiltrate computer systems via social engineering: by disguising themselves as harmless files until executed by an unknowing user.

The arms race continued with the birth of worms. Network bound worms do not rely on infected software or human factors to propagate as they are engineered to exploit network services. Scanning and infecting new victims is a task easy to automate, and consequently the propagation times are significantly faster. If utilizing an unpatched "0-day" exploit, a network worm can be released in the wild and infect millions of vulnerable systems before vendors or authorities can respond.

Viruses, trojans, and worms are usually pre-programmed with their mischievous deeds: performing their malicious duties immediately, or relying on a cue from the system in the form of time-delays or logic-bombs. While attackers could in principle gain control of their targets via back-doors, such practice was not common as it could not be scaled to many victims.

Before the 1990's, malicious software was predominantly intended to attack, destroy, and cause general mischief in an automated fashion. The next evolution of malware, the bot, changed all of this.

The paradigm of modern cybercrime is recruitment and utilization of infected computers as a strategic asset. Infected

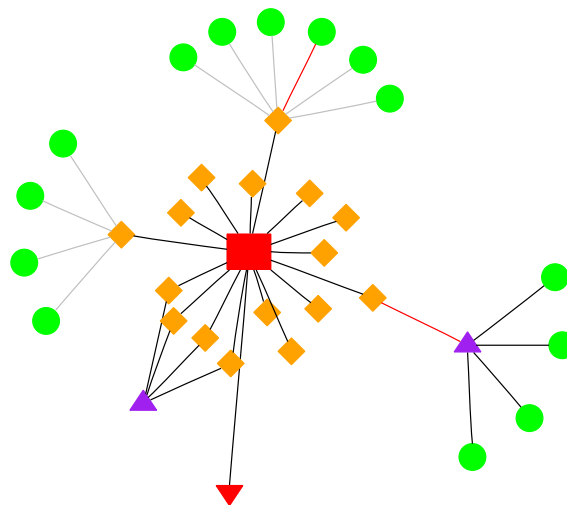


Fig. 1. A botnet with a swarm of drones. The herder (inverted triangle, bottom) logs into a centralized command and control server (square, center), where he instructs his drone army (diamonds) to scan for new victims. One of the drones finds a victim and infects it (circle, top). Additionally, several drones are used to attack a webserver (triangle, lower left) in the form of a DDoS attack.

victims are no longer disposable casualties in an attack, but a resource to be collected and built upon.

A botnet is a collection of victim computers, known as bots, drones, or zombies, that have been infected and assimilated into a greater collective through a centralized command and control (C&C) infrastructure. The simple addition of this communications back-channel enables the attacker (also bot-master, or bot-herder) to perform a variety of attacks that rely on significant computational or network resources.

Botnets can grow to millions of drones. Infected populations span the commercial and home sectors, and in some cases may even find their way into military and government networks. The wide base of infected personal computers and the large coverage amongst the world internet infrastructure gives botnet owners unprecedented power and resources in the digital age.

Botnets can be utilized for many nefarious duties:

- DDoS attacks - Network services can be incapacitated by flooding them with millions of useless requests. Bot herders exploit this vulnerability with thousands of infected drones, making the attack difficult to defend

against.

DDoS attacks have been shown to be immensely powerful against entire corporations and nations. In late April, 2007, the nation of Estonia sustained a massive distributed cyberattack that disabled critical infrastructure and media.

- Infection - Bots can scan for and infect more hapless victims with network vulnerabilities.

Increasingly, web-based infection mechanisms have been observed in the wild, whereby drone populations actively spider the web for known web application vulnerabilities. Legitimate websites become infected with a drive-by exploit capable of attacking and infecting vulnerable web-browsers. Consequently, subsequent visitors to the page can become infected. Such attacks can leverage a given website's popularity to attack most of its frequent visiting customers.

- Spamming - Botnets have risen to become a significant source of the world's email spam. Increasingly, bot herders are relying on spam-related services as a primary income source. Such spam can also bring phishing attacks and additional infections.
- Espionage - Any victim of a bot infestation becomes the target of spying. Bot software is capable of searching the local filesystem for documents to send back to the attackers, as well as extracting all saved internet passwords from Internet Explorer's protected storage systems. Many bots are additionally equipped with keylogging modules, capable of intercepting all keystrokes and window actions from within the windows API. Some bots have even been observed with features to access webcams and take full screenshots of the desktop.
- Proxies - Many bots have proxy capabilities. Utilizing these features, a criminal can effectively use his drones to route his communications and evade capture from authorities.
- Clickthrough Fraud - Many popups have been attributed to botnet-related fraud, and several herders have made small fortunes by instructing their army of drones visit websites with affiliate advertisements. Bots can also be commanded to download and install adware to earn additional profits.

Bots continue to evolve. Bot software can be downloaded for free from many online communities. New exploits and features are added continuously as an open-source collaboration between criminals. Consequently, partaking in cybercriminal endeavors is easier than ever.

A. The Underground Economy

Beneath the botnet computer networks lies the hidden social network of cybercriminals. Amongst them thrives an underground economy of vast proportions.

Entire communications networks have been located whereby spammers, bot-herders, malware authors, and other criminals gather to cooperate and sell their services [4]. Among the exchanges that occur in these economies is a vibrant market

of compromised credit cards and financial accounts. Criminals who compromise these accounts electronically are usually unable to extract the funds, so they sell them for pennies on the dollar to hardened criminals who can.

Many botnets are actually rented to other criminal organizations, usually for spamming related activities. While ad-laden spam can be annoying, phishing attacks and stock market pump-and-dump scams can bring untold losses to individuals.

These communities are now diving further and further underground, and now utilizing encryption techniques to hide from authorities who are becoming more aware of their existence [4].

B. Terrorism Online

Recently, it has been found that terrorist groups are beginning to take notice in the promise behind cybercrime.

Waseem Mughal, Younis Tsouli, and Tariq al-Daour were found by investigators in the U.S. and Britain to be engaging in various acts of cybercrime. The group had been caught utilizing stolen credit cards to establish an online network of jihadist propaganda websites and communications forums, and had also disseminated tutorials on bomb-making and computer hacking [8].

Younis Tsouli, a Moroccan native, had been an affiliate of Musab al-Zarqawi. He was known amongst the counter-terror community as Irhabi007, which translates to "terrorist-007". He is considered an early pioneer of al-Qaeda's internet propaganda efforts, reaching many potential recruits worldwide who are geographically isolated through online media.

Tariq al-Daour was arrested having possession of 37,000 [8] stolen credit card numbers, along with personal information from victims. These credit cards had been achieved through online phishing attacks against consumers, as well as trojan horses. The trio had made more than \$3.5 million in charges on these compromised accounts, and had purchased prepaid credit cards and more than 250 airline tickets [8].

The arrests of these three individuals have confirmed the applicability of cybercrime research to national security issues. It is imperative to address these concerns before additional terrorist groups make use of such tactics to gain funding or damage network infrastructure.

II. SHADOWSERVER

The ShadowServer Foundation is a nonprofit group of volunteer watchdog security professionals whose mission is to gather intelligence from the darker side of the internet. Initially specializing in botnet hunting, ShadowServer has begun to branch into related areas of electronic fraud investigations and malware research.

The team follows a rigid methodology of threat detection, analysis, investigation, and mitigation, whereby new threats are carefully considered and, if necessary, brought to the appropriate authorities to assist in incident response.

Among its many research projects, ShadowServer actively detects and monitors botnets, and it is the historical botnet logs that this study is based upon.

A. Honeypots

In computing, a honeypot is a network resource that is either vulnerable to or emulates security vulnerabilities with the express purpose of attracting attacks by cybercriminals. Honeypots are utilized to detect security threats, understand the behaviors and motivations of attackers, and to collect malicious software (hereafter known as malware).

ShadowServer makes use of malware honeypots to passively collect malware from attacking hosts online, including the freely available malware-collecting honeypot known as Nepenthes, which is developed and maintained by the mw-collect alliance [2]. The honeypot is low-interaction, in that it passively emulates known network-bound vulnerabilities to successfully extract malware samples from attacks.

Furthermore, we have made use of active honeyclient spiders, which are live windows systems which crawl the net looking for websites that engage in drive-by-downloads. When visiting a malicious website with a live vulnerable web browser, the honeyclient sustains an infection. The attacking malware is extracted from the now-tainted honeyclient and is passed on for analysis.

ShadowServer is currently investigating new technologies, including p2p-based malware harvesters to look for malicious files passed on as trojan horses on popular file-trading platforms. Furthermore, preliminary analysis of malicious files distributed on warez sites is being performed.

Having placed passive honeypots in key networks around the world, and periodically harvesting malware from known malicious sites, ShadowServer centrally gathers all malware for passive processing and active sandbox analysis.

B. Malware Analysis

1) *Passive*: Passive malware analysis involves scrutinizing each malware sample without actually executing it.

ShadowServer scans all incoming malware with many of the most popular AntiVirus engines, including Panda, AntiVir, F-Secure, Kaspersky, Norman, Vexira, VirusBuster, F-Prot6, ClamAV, NOD32, AVG, BitDefender, Avast, DrWeb, VBA32, and McAfee. Each engine's detection rate is reported on the shadowserver website [1].

ShadowServer also utilizes fuzzy-hashing [5] as an additional classification mechanism for incoming malware. Additional work is being performed to cluster and investigate the inter-relationships between different samples.

2) *Active*: Sandbox analysis is a technique which enables a malware researcher to actively execute untrustworthy malicious code in a carefully controlled and monitored environment. ShadowServer makes use of in-house sandbox software to execute malware and extract relevant network traffic over the wire.

In addition to in-house software, ShadowServer makes use of Sunbelt Software's CWSandbox [6], a commercial but powerful sandboxing solution. CWSandbox allows malware samples a limited execution time within the windows environment. System calls made to the underlying windows API are recorded and included in a detailed xml report.

All sandbox output is parsed for any relevant malware statistics. Our primary interest is whether or not the malware sample generated any network traffic, including the IRC protocols used by the majority of botnet-related malware.

C. Snooping

Any IRC servers accessed during sandbox analysis are assumed to be malicious, and thus closely monitored by ShadowServer. The team connects to and monitors newly-discovered IRC networks and records all IRC traffic. The IRC logs are analyzed by a pattern-matching signature system, whereby events and commands are classified by type. The extracted information is eventually disseminated in reports to various parties around the world, including CERT groups and vendors.

Outside of IRC communications, a small subset of botnets today are beginning to utilize HTTP and p2p communications protocols for C&C mechanisms. Any network traffic found during sandbox analysis is further scrutinized to look for new threats. The HTTP and p2p botnet scene is a fast-evolving landscape and ShadowServer is actively following the developments.

III. INVESTIGATING THE BOTNET WORLD

During the summer of 2007, a cursory analysis was performed on ShadowServer's collected botnet logfiles to investigate possible advanced data-mining practices in the future. The investigation examined the recorded events of 3,611 unique botnet C&C channels with upwards of 4000 distinct monitored individuals, spanning April 2006 to July 2007.

A prime motivation for the study was to increase our intelligence on the bot-herder underground, and the general state of the botnet threat at hand. How many herders were we dealing with? Of those we do successfully monitor, who are the most prolific amongst the community? Do these prominent criminals form gangs, and if so what are those groups engaged in? Who should law enforcement focus on first, given the severity of crimes? Who is the most dangerous?

A. Dataset Processing

The dataset consists of 20 gigabytes of raw ascii text, split amongst separate logfiles for each monitored C&C channel. The contents of each log consist of IRC protocol traffic structured with a simple "event-type" format. As IRC clients enter or exit the chat channel, JOIN and QUIT events are reported, detailing the nicknames and ip-addresses of the connecting clients. Among these nicknames are the many infected drones, as well as the criminal herders that control them.

In addition to JOIN/QUIT events, PRIVMSG events are displayed in the logs whenever a message was sent to the chat channel. In the case of IRC-based botnets, it is within these "PRIVMSG's" that most controlling commands are issued to the infected drones. Additionally, drones communicate back to the criminals via PRIVMSG's in the form of myriad status messages.

```

<E8159> :root!hackMe@333.ip JOIN #dvdripfr
<E7534> : [CHN]17!gu@60.2.12.29 JOIN :#pwn
<E8159> :slns!u@cz.org PRIVMSG #dvd :.login
<E7534> : [USA]56!ve@rs.rr.com JOIN :#pwn

```

Fig. 2. A botnet log snippet. In the JOIN messages, nicknames are connecting to channels. Lines 2 and 4 are likely connecting drones, as the nicknames look like random ID's. Line 3 is a herder issuing a login command to his drones. Due to this activity, our signature system will select "slns" as a herder nickname for the C&C id "E8159", and reject the other nicknames as drones.

Several other event-types are supported by the IRC protocol. The TOPIC keyword, for example, enables an operator of the channel to set the channels current topic of conversation. While in normal IRC chat communication this directive has a well-defined purpose, in IRC botnet communications it is often used to assign work to infected drones as they connect to the C&C server. The NICK event signifies when a client, using a particular nickname, is switching to a different nickname.

To process the contents of each logfile, a simple signature system was used. A large set of regular expression signatures have been written by ShadowServer staff over the years to scrutinize the botnet logs. Not only do these signatures recognize the IRC protocol events, but they also detect and classify commands and status messages typed by criminals or printed by infected bot software.

While the IRC protocol is well-defined, the range of botnet messages encountered in the wild is incredibly diverse. Not only can different bot software packages speak different commandsets, but the individual criminals will additionally modify their infectious payloads to avoid detection and classification.

Despite these difficulties, ShadowServer staff have assembled a large collection of signatures to classify known command strings for the various actions they signify. Commands are classified into several categories, such as "DDoS command", or "infection event", or a "password-theft event". Using this set of signatures, and a parsing engine written in Perl, the entire contents of the botnet backlog were analyzed and classified. This produced a compendium of what events transpired in each channel.

1) *Nickname Enumeration*: Amongst the noise, nicknames of channel participants can be parsed from PRIVMSG, JOIN, QUIT, and NICK events. Most nicknames encountered in C&C channels are indeed drones, however a small subset of nicknames may issue commands to operate the botnet. These are the criminals. Human nicknames must be isolated from the infected drones to perform any social network analysis.

Many bots, when connecting to their host C&C channels, will assume a random numeric ID code. This convention is not valued by all herders however, and in many cases drone nicknames will originate from a dictionary built into the malware. Thus, identifying drones through their nicknames alone can be less than trivial, and certainly not easily automatable.

The solution to the problem was to utilize the aforementioned signature system to pick out nicknames found issuing bot command strings. Drones could also be positively con-

firmed as "nonhuman" by a large set of characteristic drone status messages. These heuristics produced a sanitized list of the leaders of each and every monitored C&C.

2) *Drone Counting*: While the IRC protocol natively supports a query mechanism through which the current channel population can be found, we have found that most botnet C&C's do not report this information. Consequently, additional estimators are needed to establish a botnet's infection size. A system was devised in which nicknames of channel participants are tracked via JOIN, QUIT, PRIVMSG, and NICK events.

A simple approach can be taken whereby each nickname encountered in the channel has its state tracked in a lookup table. JOIN and QUIT events will increment and decrement a population counter depending on whether or not the nickname has been detected before. While JOIN events will always indicate a channel population increase, a QUIT event will only signify a population decrease if we saw the nickname JOIN previously in time.

A more refined approach was taken to arrive at more accurate population estimates. When a new nickname is "discovered" through any nickname-centric IRC event (PRIVMSG, NICK, JOIN, QUIT), the estimator increments the botnet population for all time prior to that event occurring. This compensates for any unlogged JOIN events that have occurred prior to our snoops connection to the channel.

To complicate matters, some C&C channels have been found that neglect reporting JOIN events, QUIT events, or both. The practice of crippling this important IRC protocol functionality is used to optimize the bandwidth requirements of hosting a C&C server, which leads to better performance and stealth. On these channels, populations can still be discovered with PRIVMSG events, however it can be impossible to decrement population sizes without information from the missing QUIT events, whereby drones will reconnect later with a new random nickname.

Populations on these exotic channels can be estimated by assuming an average time-to-live for drones, and this time-to-live can be estimated from diurnal growth modeling [7]. To simplify matters however, we merely disregarded these difficult-to-characterize channels in our drone counting.

The successfully computed drone population estimates gave us a rough measure for the infection base of each monitored channel, as well as an evolution curve for the population of each botnet (Figure 3). The results also confirmed the diurnal population hypothesis discussed in previous literature [7]. The population size of a botnet oscillates with a 24-hour period as some fractional subset of victims will indeed power down their computers at night.

B. Key Players

The simplest metric by which we can gauge an individuals prevelance in the botnet community would be to count the number of C&C channels they were observed participating in. The results provide a staggering look into the assets controlled by some of these criminals.

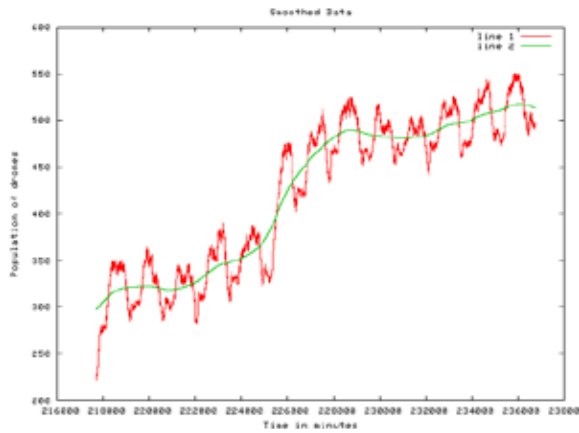


Fig. 3. Results of botnet population estimation. The botnet in question is experiencing diurnal activity patterns and linear infectious growth. The fit line is the result of gaussian-kernel based smoothing with a full width half max of 24 hours.

nickname	# C&C's	nickname	# C&C's
always	65	aaa	40
joeblow	60	evil	39
emr3	60	MArian0z	36
KoRn	59	gu3sT	33
lol	58	CeeK	33
asd	57	Busy	33
Marvin_	53	Source	31
bill	52	JuMp	31
root	51	aa	31
process	48	Lindi_Cracker	30
hidden	47	mjd	29
xx	46	KaHiN	29
mr'bet—	42	as	29
edzy	40	tonii	28
D—_PaLo	40	ILGuardiano	28

Fig. 4. Top 30 most prolific botnet herders

While this simple analysis gives insight into who is potentially the most successful of the botnet herders by counting their controlled C&C assets, it fails to explore any inherent structure found amongst entire communities. Furthermore, these statistics may emphasize nicknames found to be infiltrating and stealing bots from other criminals. Such practice has become common, as many herders are skilled enough to detect other's botnet C&C channels and subvert their security mechanisms.

C. Criminal Social Network

The aforementioned signature system enables tagging of specific crimes to specific channels and, in some cases, to individual nicknames. While the enumeration of these threats is useful in dealing with specific incidents as they occur, they fail to illuminate the long-term motivations of entire groups of criminals who may be collaborating with each other amongst many groups of channels.

Analysis was performed on the community structure of the underlying social network. All pre-filtered "human" nicknames

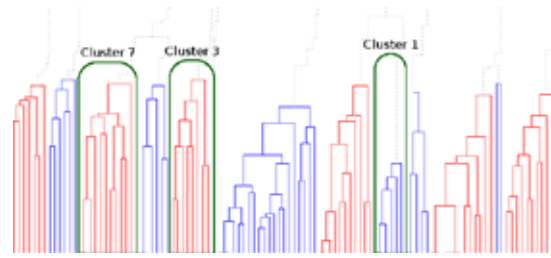


Fig. 5. A small component of the hierarchical clustering results showing the inherent staggering of group participation in three of the discussed gangs. Some members are indeed more involved in operating their collective assets than others.

were taken as vertices in a large undirected graph. Edges were defined between any two nodes found collaborating in a single C&C channel. Weights were assigned to the edges with a simple Jaccard metric measuring the percentage of channels the nicknames shared in common divided by the total number of channels occupied by either nickname. That is, the weight $w_{(ij)} = \frac{|T_i \cap T_j|}{|T_i \cup T_j|}$, where T_n is the set of C&C channels that nickname n was found occupying.

A hierarchical agglomerative clustering algorithm was used to cluster vertices together based under an average group linkage heuristic requiring a minimum similarity of 50%. The resulting dendrogram (Figure 5) has 104 clusters containing 957 nicknames.

For each resulting cluster of herders, a list of associated C&C channels was assembled based on the nicknames contained within. From this joining, the combined criminal events for each cluster can be taken in aggregate to reveal a groups overall criminal focus.

Of the 104 clusters, a small subset of significant groups is listed in table (Figure 6), along with a summary of the criminal activity detected by the aforementioned signature and bot-counting systems. The estimated number of controlled drones is listed, as is both the number of unique DDoS targets victimized and the number of passwords stolen from unsuspecting users.

Overall the results show some of the largest gangs personally observed by the author to be prominent in the botnet scene. While the first 5 groups were well known by the author, the last 3 shown, as well as some 20 other groups not listed, were brought to light as potential future targets of significant investigative scrutiny.

IV. FURTHER WORK

While the analysis techniques outlined above are an excellent start for obtaining a big-picture look at the botnet underground, several additional steps must be taken to better filter and condition the data. Furthermore, additional investigation is needed to better scrutinize the motivations behind the logged crimes.

- Many herders will use close variations of a similar nicknames. (For example, "process" and "process-away"). In most cases it is quite obvious that it is the same individual

Herders	C	D	B	P
[0]USA—2KSP3[Om]824584, creature, edzy, fri, frioz, wejbwfe, wloo, BlaCkD3v—L	51	1263	235713	
bill gu3sT Besi D—_PaLo hidden load process tonii	88	3310	30140	
Albania DaddyCool[a] jelo jeloo [KleviS] Opium Silv3r-ArRoW waleed	44	730	252969	
ILGuardiano liga MARIanOz PepP0z JuMp	56	6698	256193	
xRaZoRx xBreaKx xxDCxx vDCv xGoDx xCKx xBeNx xBrandoNx xAmplifyx xSKYx xToaDx xTiMx	15	2350	6094	1988
Max hans matrix toxic abc Peter bob home Andy dan Jack blbla billy mark xxx sss mr	15	2615	303286	
StRuGaNi_007 bostss Heropos niggaz yeste Pacino NhG Ld fada pilz AsC [a] bAcaRdl dRiVeR alejandro mut hook Dritton ArditS Corrupted	32	730	220703	
Attacker hh	12	479	239708	4484

Fig. 6. Several representative interesting groups found, based on concentrations of criminal events and membership size. The columns, from left to right, represent the member nicknames, the number of C&C channels controlled by the gang (C), the number of unique DDoS targets (D), estimated number of bots (B), and number of PSTORE compromises of victim passwords (P).

sprawling multiple channels. In this investigation, obvious nickname duplicates were corrected by hand (such as Lupin3 and LupinIII), however such a data conditioning effort cannot be reliably automated.

While the IRC "NICK" event can be used to follow nickname changes over time, some special circumstances warrant careful scrutiny. One group of decidedly distinct individuals was found to be using eachothers nicknames frequently, either as a joke or to evade surveillance. In another channel, some short nicknames (like "x" or "a") were found to be "hubs" to other nicknames in the NICK graph. It was later determined that these popular nicknames were IRC bouncers used to anonymously proxy connections. Such practices observed in the wild make it difficult to both ascertain the identity or even existence of multiple individuals. Additional work is needed to profile behavioral characteristics of herders.

- While hierarchical clustering proved successful in automatically grouping criminals together who were observed to be co-conspirators, it overall fails to segregate the IRC channels themselves. An overall better approach would be to apply biclustering or hypergraph partitioning to the bipartite graph of nicknames and channels. Such an approach would best resolve the overall staggered community structure found in this investigation.
- To better profile the DDoS attack motivations of these gangs, the actual DDoS targets must be individually scrutinized. Current work is in progress to quantify the distribution of targets amongst national network ranges.

Additionally, IP addresses could be correlated with latitude and longitude to determine any geopolitical or regional focus.

Ideally, attacked websites should be automatically text-mined to determine the subject matter. If correlations between many attacked sites are found over a given time period, coordinated hacktivism could potentially be automatically detected and tied to the correct gangs.

V. CONCLUSION

The Cyber Underground has dramatically evolved in recent years to encompass world-influencing crimes with large sums of money at stake. Some prolific players in the botnet scene have built, upon the resources of countless infected and unwilling drone machines, computerized armies with an unbelievable amount of power. Behind these spreading networks is a community of cybercriminals engaging in serious crimes that influence people, markets, and entire nations.

As these communities dive deeper underground and as encryption technologies become more widespread, tracking these miscreants and their botnets will become more and more challenging. While public knowledge of botnets and electronic fraud increases, the problem of mass-infection of vulnerable networked computers and ubiquitous cybercrime will not be vanishing. The only defense we have against this new wave of cybercrime is for individuals to secure themselves at the personal level, and for networks to secure against mass-infection.

The time is NOW to act on the compounding botnet problem. To this date, the networks we monitor continue to grow and the underground economy still thrives. With greater numbers of honeypot sensors, more monitoring of detected threats, and greater cooperation with authorities, ShadowServer hopes to assist in whatever way we can to make the internet a safer place.

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